Score-Aligned Reward Design for Deployable Deep Reinforcement Learning Control of Community Battery Storage

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***Abstract*—Batteries in grid-interactive buildings can lower electricity bills and emissions by shifting consumption in time, but a controller that runs in a real building never knows future prices, carbon intensity or demand. Model-free deep reinforcement learning (RL) is attractive because it learns a control policy from interaction alone, yet its performance depends heavily on how the reward encodes the operator's objectives. We present a systematic study of 52 reward, algorithm and input designs for district-level battery control on the five-building CityLearn Challenge 2022 benchmark, using only genuinely deployable RL: every policy acts on current measurements, past information and the forecasts provided by the environment, and is trained from reward alone. The key ingredient is a *score-aligned counterfactual reward*: per-step terms computed exactly as the official district-level key performance indicators (KPIs), scaled by their no-battery values, weighted like the challenge score with an increased carbon weight, and expressed as the improvement over doing nothing in the same hour. With Soft Actor-Critic (SAC) it lowers electricity cost by 28.0% and carbon emissions by 18.3% relative to no battery over three seeds (Phase I score 0.768 ± 0.005), whereas a rule-based controller and a PPO cost-only agent are worse than or equal to not using the batteries (1.132 and 1.004). With Truncated Quantile Critics (TQC) the same reward gives the best overall challenge score (0.861 ± 0.003), and TQC with features built only from past observations gives the best Phase I score (0.763 ± 0.005; cost -29.1%, emissions -18.4%). These policies, unchanged, transfer to five unseen buildings with cost and carbon reductions of 29.1% and 20.9%. Ablations show that off-policy learning, the counterfactual baseline, KPI-exact terms and carbon weighting each contribute, whereas larger networks, CrossQ, multi-step returns and longer training do not help.**

***Index Terms*—Battery energy storage, demand response, deep reinforcement learning, soft actor-critic, reward design, CityLearn, carbon-aware control, grid-interactive buildings.**

I. Introduction

BUILDINGS account for a large share of electricity demand, and distributed photovoltaic (PV) generation and behind-the-meter batteries turn them into flexible grid resources. By charging when electricity is cheap or clean and discharging when it is expensive or carbon-intensive, a battery can reduce both the bill of its owner and the emissions caused by its consumption, while a coordinated fleet can also smooth the aggregate load seen by the distribution grid [1], [2].

Classical control of such systems relies either on hand-crafted rules or on model predictive control with forecasts of demand, generation, prices and carbon intensity. Rules are simple but cannot adapt, and optimisation-based control requires accurate models and forecasts that are costly to build for every building. Model-free deep reinforcement learning (RL) [3] offers an alternative: a policy is learned directly from interaction with the system, without an explicit model, and acts in real time on the information available at each step [4], [5]. The CityLearn environment [6], [7] and the NeurIPS 2022 CityLearn Challenge [8] made this setting a common benchmark, with an official score that combines electricity cost, carbon emissions and grid KPIs.

A recurring difficulty is that the quantities an operator cares about are yearly, district-level KPIs, whereas an RL agent optimises a per-step scalar reward. Naive choices, such as the negative bill of each building or a weighted sum of terms on arbitrary scales, are poorly aligned with the official KPIs: they ignore that exports earn nothing in the official cost, that buildings offset each other at district level, and that terms with different magnitudes dominate one another. They also produce high-variance learning targets because most of the reward is driven by exogenous demand rather than by the battery action. Our experiments show that such misalignment can make learned controllers no better than leaving the batteries idle.

This paper studies how far genuinely deployable model-free RL can go on the CityLearn 2022 benchmark when the reward is designed carefully. We deliberately restrict ourselves to controllers that could run in a real building: the policy sees only current and past measurements and the forecasts that the environment itself provides, and it is trained from reward alone, without access to future data or to precomputed optimal schedules. Our contributions are:

1) A systematic, multi-seed comparison of 52 RL controllers spanning PPO, SAC, TQC and CrossQ, central and decentralised agents, eight reward families (including fixed and adaptive multi-KPI weighting), several network sizes, past-only history features and training lengths.
2) A *score-aligned counterfactual reward* whose per-step terms sum to the official KPIs, scaled by the no-battery reference and expressed as the improvement over idling; with SAC it reduces cost by 28.0% and emissions by 18.3% relative to no battery; with TQC it gives the best overall score, and with TQC and past-only history features the best cost, carbon and Phase I score.
3) Ablations that isolate the effect of each design choice, and negative results (reward normalisation, decentralised agents, residual learning on a rule-based controller, larger networks, longer training) that are useful to practitioners.

4) A generalisation test on five unseen buildings, where the learned policies keep their cost and carbon reductions without retraining.

The remainder of the paper is organised as follows. Section II surveys related work. Section III formulates the control problem and the KPIs. Section IV describes the agents and reward designs, Section V the experimental protocol, and Section VI the results. Section VII discusses limitations and Section IX concludes.

II. Related Work

A. Reinforcement Learning for Building Energy and Demand Response

Vázquez-Canteli and Nagy [1] reviewed RL for demand response and found that most studies used tabular or simple function approximators, single buildings and bespoke simulators, which made results hard to compare. Wang and Hong [2] and Yu *et al.* [4] surveyed deep RL for building control, identifying sample efficiency, safety, reward design and generalisation across buildings as the main open problems. Nagy *et al.* [5] discussed ten questions on RL for building energy management, including the need for standard benchmarks and KPIs. Early deep RL applications include HVAC control with deep Q-networks [9], [10] and on-line building energy optimisation with deep policy-gradient and Q-learning methods [11]. Dulac-Arnold *et al.* [12] formalised the challenges of applying RL in the real world, many of which (multi-objective rewards, partial observability, limited data) are central to battery control.

B. CityLearn and Grid-Interactive Communities

CityLearn [6] is an OpenAI Gym environment for coordinated demand response in districts of buildings with storage and PV; CityLearn v2 [7] extends it with electric vehicles, occupant comfort, resilience and carbon-aware control. MARLISA [13] coordinated multiple agents through iterative sequential action selection for load shaping. Nweye *et al.* [14] analysed real-world challenges for multi-agent RL in grid-interactive buildings, and MERLIN [15] studied offline and transfer learning for occupant-centric operation of communities. The NeurIPS 2022 CityLearn Challenge [8] evaluated controllers on cost, carbon and grid KPIs normalised by the case without batteries; we use its dataset, its official KPI definitions and its public leaderboard as reference points.

C. Deep RL Algorithms for Continuous Control

Battery set-points are continuous, so actor-critic methods are the natural choice. DDPG [16] and TD3 [17] learn deterministic policies with off-policy critics, TD3 adding clipped double Q-learning to curb over-estimation. Soft Actor-Critic (SAC) [18], [19] maximises reward plus policy entropy, with automatic temperature tuning, and is among the most robust off-policy methods. Proximal Policy Optimization (PPO) [20] is a widely used on-policy alternative. More recent work improves sample efficiency and value estimation: TQC [21] truncates a mixture of distributional critics, REDQ [22] uses an ensemble of critics with a high update-to-data ratio, CrossQ [23] removes target networks through batch normalisation, and multi-step returns [24] propagate delayed rewards faster. We use the Stable-Baselines3 implementations [25].

D. Reward Design and Multi-Objective Weighting

Battery control is inherently multi-objective [26]: cost, emissions, ramping and peaks often conflict. Scalarising them requires weights and scales; poorly scaled terms let one objective dominate. Ng *et al.* [27] showed which reward transformations leave the optimal policy unchanged, a property we exploit with an action-independent counterfactual baseline. In multi-task learning, Dynamic Weight Average [28] and uncertainty weighting [29] adapt loss weights during training; we adapt the former to KPI weights in our adaptive multi-KPI rewards. For storage arbitrage specifically, Cao *et al.* [30] showed that including an accurate battery degradation model in the reward changes the learned arbitrage strategy, motivating the small battery-wear term in our rewards.

E. Evaluation of RL Results

Henderson *et al.* [31] and Agarwal *et al.* [32] showed that deep RL results vary strongly with random seeds and that single-run comparisons are unreliable. We therefore report every result as the mean and standard deviation over three independent seeds and evaluate deterministic policies over a full year.

III. Problem Formulation

A. Environment

We use CityLearn 2.1.2 with the `citylearn_challenge_2022_phase_1` dataset: five residential buildings with rooftop PV and one battery each, simulated hourly for one year (8,759 control steps). Each building has an uncontrollable non-shiftable load and PV generation; the controller decides, every hour, the charge or discharge rate of each battery as a fraction of its capacity. Electricity is billed with a time-of-use tariff and the grid has a time-varying carbon intensity.

B. Observations and Actions

A central agent observes 60 values: calendar features (month, day type, hour); outdoor temperature, relative humidity and diffuse and direct solar irradiance, each with the environment's 6, 12 and 24-hour forecasts; the current carbon intensity; and, for each building, its non-shiftable load, PV generation, battery state of charge (SoC), net electricity consumption, and the current electricity price with its 6, 12 and 24-hour values from the tariff. The action is a vector of five continuous battery rates. No future demand, generation or carbon intensity beyond these provided forecasts is available to the policy, so every controller in this paper can be deployed as is.

C. Key Performance Indicators

Let E_t be the district net consumption at hour t (positive for imports), π_t the price and I_t the carbon intensity, and let a superscript "nb" denote the same quantity in the reference run without batteries. The official challenge KPIs are

$$C = \frac{\sum_t \max(0, E_t \pi_t)}{\sum_t \max(0, E_t^{\mathrm{nb}} \pi_t)}, \qquad G = \frac{\sum_t \max(0, E_t I_t)}{\sum_t \max(0, E_t^{\mathrm{nb}} I_t)}, \quad (1)$$

$$R = \frac{\sum_t |E_t - E_{t-1}|}{\sum_t |E_t^{\mathrm{nb}} - E_{t-1}^{\mathrm{nb}}|}, \qquad D = \tfrac{1}{2}\big(R + (1 - L)\big), \quad (2)$$

where $1 - L$ is the average monthly $1 - \bar{E}/\max E$ relative to no battery. Exports earn nothing, and because the clipping is applied to the district total, one building's export offsets another's import. All KPIs are ratios to no battery, so values below one are improvements. We report the Phase I score $(C + G)/2$, used for the challenge's Phase I leaderboard on these five buildings, and the score $S = (C + G + D)/3$, the challenge's overall formula.

IV. Methods

A. Agents

Central agents control all five batteries with one policy. *Decentralised* agents share one policy across buildings (parameter sharing): each building observes its own values plus a one-hot building identifier and outputs its own action. We train PPO [20] and SAC [18] agents with Stable-Baselines3 [25], and TQC [21] and CrossQ [23] agents with its sb3-contrib extension. We also test a residual agent whose action is added to the action of the rule-based controller (RBC) supplied with CityLearn, and variants with running normalisation of observations and rewards.

B. Reward Families

We compare eight reward families (Table I): (i) the negative cost of each building; (ii) cost plus a static carbon weight; (iii) cost plus a carbon weight proportional to the current carbon intensity, or active only in the dirtiest 25% of hours; (iv) a multi-KPI reward

$$r_t = -\big(\hat{c}_t + \mu_g \hat{g}_t + \mu_r \hat{r}_t + \mu_p \hat{p}_t + w_b \hat{b}_t\big), \quad (3)$$

whose terms are cost, carbon, ramping $|E_t - E_{t-1}|$, the increase of the daily peak, and battery wear $\sum_i |\mathrm{SoC}_{i,t} - \mathrm{SoC}_{i,t-1}|$, each divided by its per-step mean under the RBC; (v) the same with *adaptive* weights μ, updated once per simulated week in the style of Dynamic Weight Average [28]: the weight moves toward the KPI furthest behind its target, with a floor so that no KPI is ignored and a fixed total; (vi) cost and carbon plus a squared-load penalty; (vii) a *counterfactual* multi-KPI reward that adds back the same weighted terms of the no-battery run in the same hour; and (viii) the *score-aligned* rewards described next.

C. Score-Aligned Counterfactual Reward

The proposed reward makes the per-step terms add up exactly to the numerators of the official KPIs and puts them on the scale of the KPIs. At hour t we use the district-level clipped cost and emissions,

$$\hat{c}_t = \frac{\max(0, E_t \pi_t)}{\bar{c}^{\,\mathrm{nb}}}, \qquad \hat{g}_t = \frac{\max(0, E_t I_t)}{\bar{g}^{\,\mathrm{nb}}}, \qquad \hat{r}_t = \frac{|E_t - E_{t-1}|}{\bar{r}^{\,\mathrm{nb}}}, \quad (4)$$

where a bar denotes the per-step mean of the term in the no-battery run, so that $\sum_t \hat{c}_t = T \cdot C$: the yearly return is a linear function of the official KPIs. The peak term $\hat{p}_t$ is the increase of the running daily maximum of E_t, with the first hour of each day counting its full load, so that a day's terms sum to that day's peak (an earlier version without this correction let the agent hide a peak in the first hour of the day). The reward is

$$r_t = -\big(\hat{c}_t + w_g \hat{g}_t + w_r \hat{r}_t + w_p \hat{p}_t + w_b \hat{b}_t\big) + \big(\hat{c}_t^{\,\mathrm{nb}} + w_g \hat{g}_t^{\,\mathrm{nb}} + w_r \hat{r}_t^{\,\mathrm{nb}} + w_p \hat{p}_t^{\,\mathrm{nb}}\big). \quad (5)$$

The second bracket is the value of the same terms had the batteries stayed idle in that hour. It depends only on exogenous data, not on the actions or the battery state, so it leaves the optimal policy unchanged [27], but it removes most of the demand-driven variation from the reward: the agent is rewarded for the *improvement* it causes. In a deployed system this baseline is computable in real time from the measured net load and the battery's own power. The weights mirror the score $S = (C + G + (R + 1 - L)/2)/3$: cost weight 1, ramping and peak 0.5 each (the peak term serves as a per-step proxy for the load factor), and battery wear $w_b = 0.1$. We study three weightings: *score-aligned* ($w_g = 1$), *Phase I-aligned* ($w_g = 1$, $w_r = w_p = 0.2$) and *score-aligned with carbon weight 2* ($w_g = 2$), which places extra emphasis on emissions.

V. Experimental Setup

Every controller is trained for 200,000 environment steps (about 22.8 simulated years) and then evaluated with its deterministic policy over the full year. Each configuration is run with three seeds (42, 100, 2026), and we report mean $\pm$ standard deviation. SAC uses a learning rate of $3 \cdot 10^{-4}$, a replay buffer of 200,000 transitions, batch size 256, $\tau = 0.005$, $\gamma = 0.99$, one gradient step per environment step, 1,000 random warm-up steps, automatic entropy tuning and two hidden layers of 64 units. TQC uses the same settings, with two critics of 25 quantiles each and the top two quantiles of each critic dropped; CrossQ uses a learning rate of 10^{-3}, no target networks and its default policy delay of 3. PPO uses rollouts of 2,048 steps, batch size 128, 10 epochs, $\gamma = 0.99$, GAE $\lambda = 0.95$, clip range 0.2 and two hidden layers of 64 units. The references are no battery (all KPIs equal to one) and CityLearn's `BasicRBC`. To test generalisation, the trained central policies are run without any retraining on the five unseen buildings of `citylearn_challenge_2022_phase_2`, which share the observation and action layout, and the KPIs are computed against the no-battery run on those buildings.

TABLE I
OFFICIAL KPIS RELATIVE TO NO BATTERY (LOWER IS BETTER), MEAN OVER THREE SEEDS (± STANDARD DEVIATION FOR THE TWO SCORES). BEST VALUE OF EACH COLUMN IN BOLD.

ID	Controller	Phase I	Score S	C	G	D
–	No battery	1.000	1.000	1.000	1.000	**1.000**
B0	RBC (CityLearn)	1.132	1.113	1.070	1.193	1.075
B1	PPO, cost only	1.004±.004	1.174±.015	0.939	1.070	1.515
B3	PPO, static carbon	0.984±.022	1.144±.032	0.918	1.050	1.464
B4	PPO, dynamic carbon	1.007±.068	1.181±.088	0.942	1.072	1.530
M2	PPO, multi-KPI fixed	0.959±.034	1.068±.051	0.937	0.981	1.285
M3	PPO, multi-KPI adaptive	0.950±.023	1.077±.026	0.923	0.978	1.330
M9	PPO, cost + carbon + load^2	0.900±.004	1.061±.013	0.833	0.967	1.382
M5	PPO, decentralised adaptive	1.001±.000	1.003±.000	1.001	1.001	1.006
M12	SAC, multi-KPI fixed	0.881±.020	0.982±.032	0.831	0.931	1.184
M13	SAC, + obs. and reward norm.	0.966±.053	1.004±.024	0.952	0.980	1.081
M16	SAC, residual on RBC	0.956±.026	1.007±.014	0.933	0.978	1.111
M19	SAC, multi-KPI, carbon 2	0.890±.017	0.981±.022	0.841	0.940	1.163
M20	SAC, counterfactual	0.829±.003	0.917±.009	0.781	0.877	1.091
M26	SAC, score-aligned	0.786±.008	0.870±.006	0.732	0.840	1.036
M28	SAC, Phase I-aligned	0.779±.003	0.876±.001	0.727	0.831	1.071
M27	SAC, score-aligned, carbon 2	0.768±.005	0.867±.005	0.720	0.817	1.064
M110	TQC, score-aligned, carbon 2	0.768±.004	**0.861**±.003	0.720	**0.816**	1.046
M113	TQC + history features	**0.763**±.005	0.862±.006	**0.709**	**0.816**	1.060

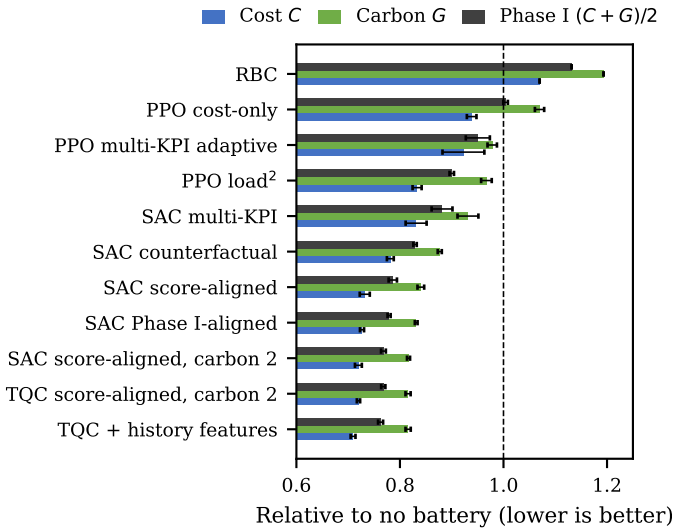


Fig. 1. Cost C, carbon G and Phase I score of representative controllers relative to no battery (dashed line); error bars show the standard deviation over three seeds.

VI. RESULTS

A. Overall Comparison

Table I and Fig. 1 summarise the official KPIs of all controllers. The RBC is worse than leaving the batteries idle on every KPI (Phase I 1.132): it charges and discharges on a fixed schedule that ignores the tariff, the carbon intensity and PV. PPO with the per-building cost or carbon-weighted rewards reduces cost by 6–8% but *increases* emissions and grid ramping (Phase I 0.98–1.01, $D \approx 1.5$), so it is no better than idling. Multi-KPI rewards help PPO (0.950 with adaptive weights), and a squared-load penalty gives the best PPO result (0.900).

Switching to SAC is the largest single step: with the same fixed multi-KPI reward, Phase I falls from 0.959 (PPO, M2) to 0.881 (SAC, M12). The counterfactual baseline then lowers it to 0.829 (M20), and making the terms exactly the official, district-level, no-battery-scaled ones lowers it to 0.786 (M26). Doubling the carbon weight gives the best SAC controller, M27, with Phase I 0.768 ± 0.005 and score 0.867 ± 0.005: cost -28.0% and emissions -18.3% relative to no battery, and a 22% lower Phase I score than the best reward of the earlier carbon-weighting baselines (B3, 0.984). Replacing SAC by TQC [21] with otherwise identical settings (M110) keeps the Phase I score (0.768 ± 0.004) and lowers ramping, giving the best overall score of the study, 0.861 ± 0.003, with the smallest spread between seeds. Adding the past-only history features of Section VI-F to TQC (M113) gives the best Phase I score, 0.763 ± 0.005, with cost -29.1% and emissions -18.4% relative to no battery.

B. The Three Best Controllers

The three score-aligned SAC controllers are the best SAC controllers of the study and are remarkably consistent across seeds (Table II): the standard deviation of the Phase I score is at most 0.008. Raising the carbon weight from 1 (M26) to 2 (M27) lowers emissions from 0.840 to 0.817 and, because cheap hours are often also clean, cost from 0.732 to 0.720, at the price of more ramping (R 1.101 to 1.156). Lowering the grid weights (M28) gives an intermediate result. All three improve the load factor ($1 - L < 1$) but increase ramping relative to no battery, so their grid KPI D stays slightly above one.

TABLE II
THE THREE BEST CONTROLLERS: KPIS (MEAN ± STD OVER SEEDS 42, 100, 2026).

KPI	M27 (carbon 2)	M28 (Phase I-aligned)	M26 (score-aligned)
Cost C	**0.720**±0.006	0.727±0.004	0.732±0.010
Carbon G	**0.817**±0.003	0.831±0.003	0.840±0.007
Ramping R	1.156	1.167	**1.101**
$1 - L$	0.972	0.975	**0.971**
Grid D	1.064	1.071	**1.036**
Phase I $(C + G)/2$	**0.768**±0.005	0.779±0.003	0.786±0.008
Score S	**0.867**±0.005	0.876±0.001	0.870±0.006

C. Ablations and Negative Results

Off-policy learning. With identical rewards SAC beats PPO by a wide margin (M2 vs. M12), consistent with its better sample efficiency on long-horizon continuous control; 200,000 steps correspond to only about 23 simulated years.

Counterfactual baseline. Adding back the no-battery terms (M12 to M20) lowers Phase I from 0.881 to 0.829 and reduces the seed-to-seed spread from 0.020 to 0.003. The baseline does not change the optimal policy, so this gain comes entirely from lower-variance critic targets.

KPI-exact terms. Replacing per-building bills by district-level clipped cost and emissions and scaling by the no-battery reference (M20 to M26) gives a further 0.043. The per-building bill rewards exports that the official cost does not, and ignores that buildings offset each other.

Normalisation hurts. Running normalisation of observations and rewards (M13) worsens SAC from 0.881 to 0.966 and

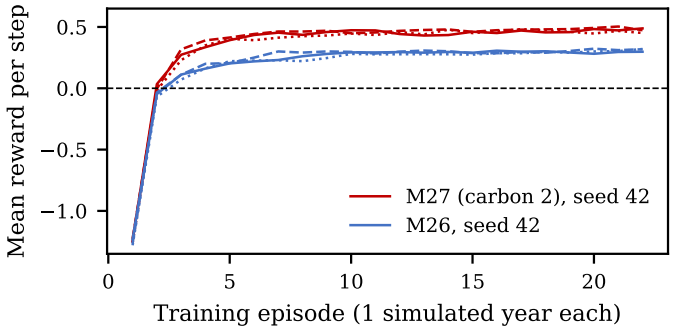


Fig. 2. Mean per-step training reward (weighted KPI improvement over no battery; 0 = as good as idling) of M27 and M26, three seeds each (solid: 42, dashed: 100, dotted: 2026).

makes it far less stable across seeds; normalising only observations (M17, 0.886) is neutral.

Decentralised PPO collapses. Shared per-building PPO policies with adaptive weights (M5, M6) converge to leaving the batteries idle (all KPIs 1.00), and adding district-level penalties (M10, M11) does not prevent it. A central SAC agent with the counterfactual reward, carbon weight 2 and per-building (non-official) cost and carbon terms also collapsed to idling (M21, one seed), which motivated the KPI-exact terms of M26–M28.

Residual RL. Learning a correction on top of the RBC (M16, 0.956) is worse than learning from scratch, because the RBC itself is a poor starting point.

D. Learning Dynamics and Learned Behaviour

With the counterfactual reward of (5), the mean reward per step is the weighted KPI improvement over idling, so zero means "as good as no battery". Fig. 2 shows it for the training episodes of M27 and M26. Both start far below zero (random exploration in the first year), cross zero in the second episode and approach their final level after about six to ten episodes; the three seeds of each method follow almost the same curve. The higher plateau of M27 reflects its doubled carbon weight, which scales its reward, rather than a better policy per se. Fig. 3 shows one summer week of the learned policy: the batteries charge from about 09:00 to 14:00, when PV covers the load, and discharge from the late afternoon through the evening and night, covering the tariff peak (16:00–21:00) and the hours of highest carbon intensity, which lowers the evening import peak.

E. Generalisation to Unseen Buildings

Table III reports the trained policies, unchanged, on the five unseen buildings of the challenge's phase 2 dataset. The score-aligned controllers keep their benefit: M27 reduces cost by 27.8% (28.0% on the training buildings) and emissions by 20.7% (18.3%), while the earlier SAC multi-KPI controller (M12) transfers much less well. The ranking on the unseen buildings follows the ranking on the training buildings, so the gains from reward design are not an artefact of fitting the five training buildings. The public formula of the challenge's Phase III, $0.4\,S_{\text{train}} + 0.6\,S_{\text{unseen}}$, gives 0.859 ± 0.005 for M27, 0.853 ± 0.002 for the TQC controller M110 and 0.854 ± 0.001 for TQC with history features (M113). M113 also gives the largest savings on the unseen buildings: cost -29.1% and emissions -20.9%.

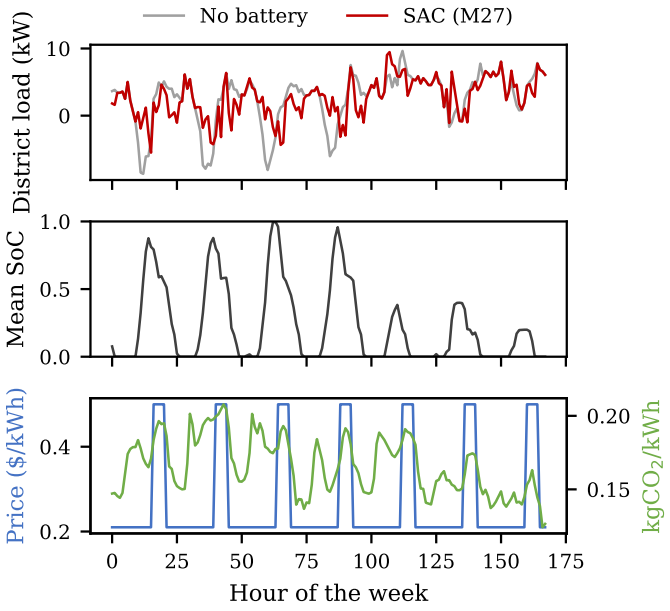


Fig. 3. One July week of M27 (seed 42): district net load with and without batteries, mean state of charge, and the price and carbon intensity the policy observes.

TABLE III
POLICIES TRAINED ON THE PHASE 1 BUILDINGS, EVALUATED WITHOUT RETRAINING ON THE FIVE UNSEEN PHASE 2 BUILDINGS (MEAN OVER THREE SEEDS; M12 AND M20: SEED 42 ONLY).

Controller	C	G	D	S_{unseen}	$0.4S_{\text{train}} + 0.6S_{\text{unseen}}$
M12 SAC multi-KPI	0.811	0.893	1.099	0.934	0.946
M20 SAC counterfactual	0.777	0.848	1.066	0.897	0.905
M26 SAC score-aligned	0.736	0.815	**1.021**	0.857	0.862
M28 SAC Phase I-aligned	0.723	0.809	1.050	0.861	0.867
M27 SAC score-aligned, c. 2	0.722	0.793	1.049	0.854	0.859
M112 M27 + history features	0.715	0.803	1.025	**0.847**	0.854
M110 TQC score-aligned, c. 2	0.718	0.793	1.032	**0.847**	**0.853**
M113 TQC + history features	**0.709**	**0.791**	1.046	0.849	0.854

F. Additional Algorithms, Architectures and Features

Starting from M27, we screened further changes, one at a time, on a shorter budget of 60,000 steps (about seven simulated years) and seed 42 (Table IV). The reference is M27 retrained with the same budget (M102), whose Phase I score over three seeds is 0.794 ± 0.016 (seed 42: 0.803). The algorithms are TQC [21], with truncated distributional critics, and CrossQ [23], with batch-normalised critics and no target networks. The history features, computed only from past observations, are the cyclic hour and month and a persistence forecast taken from yesterday: the carbon intensity and the district load minus PV $24 - h$ hours earlier for the next $h = 1, 2, 3, 6, 12$ hours. The environment provides no forecast of carbon intensity or load, and both follow a daily pattern. The "rich" variant adds each building's own lagged load minus PV, the values 24 hours earlier and yesterday's mean and

TABLE IV
SCREENING OF ADDITIONAL CHANGES TO M27 AT 60,000 ENVIRONMENT STEPS (PHASE I AND SCORE RELATIVE TO NO BATTERY; SEED 42 UNLESS STATED). CONTROL: M27 AT THE SAME BUDGET.

ID	Change from M27	Phase I	Score S	C	G
M102	none (control), 3 seeds	0.794±.016	0.882±.006	0.751	0.838
M93	history features, 3 seeds	0.808±.030	0.887±.010	0.769	0.847
M93	history features (seed 42)	0.774	0.876	0.728	0.821
M105	history + observation norm.	0.787	0.886	0.738	0.835
M107	history, batch size 512	0.787	0.878	0.744	0.830
M103	rich history	0.789	0.876	0.746	0.832
M91	TQC, 256×256	0.794	0.887	0.750	0.838
M104	dense history	0.801	0.879	0.765	0.837
M96	256×256 + history	0.801	0.884	0.755	0.847
M92	CrossQ, 256×256	0.805	0.892	0.768	0.842
M95	$\gamma = 0.95$	0.808	0.887	0.761	0.854
M90	256×256 networks	0.809	0.899	0.764	0.853
M99	2 gradient steps per env step	0.809	0.898	0.772	0.847
M97	TQC + history	0.810	0.897	0.764	0.856
M106	history, learning rate 10^{-3}	0.810	0.898	0.768	0.852
M100	CrossQ + history	0.812	0.890	0.779	0.846
M101	carbon weight 3	0.813	0.892	0.767	0.860
M108	history, $\tau = 0.02$	0.859	0.931	0.829	0.890
M94	3-step returns	0.864	0.928	0.824	0.903
M98	shared per-building policy	1.000	1.000	1.000	1.000

maximum; the "dense" variant uses every next hour from 1 to 12.

No change improved on the control by more than its seed-to-seed spread. The history features gave the best single run (0.774), but over three seeds they scored 0.808 ± 0.030, no better than the control; TQC (0.794) and the history variants with observation normalisation (0.787), a batch of 512 (0.787) or rich features (0.789) were also within one standard deviation. Larger networks, two gradient updates per step, a carbon weight of 3, $\gamma = 0.95$ and CrossQ were slightly worse. Three-step returns, a faster target update ($\tau = 0.02$) and a shared per-building SAC policy were clearly worse; the latter collapsed to idling, like the decentralised PPO agents. Averaging the actions of the three M27 seeds also hurt (Phase I 0.841), because the averaged actions are too cautious. With the reward fixed, the choice of off-policy algorithm, network size and input features matters much less than the reward design and the training length.

G. Full-Budget Tests and Training Length

Because screening at 60,000 steps is noisy, the most promising variants were retrained at the full budget with three seeds (Table V). TQC (M110), the best algorithm in the screening, matches M27 on cost and carbon and improves the grid KPI (D 1.046 vs. 1.064), giving the best score, 0.861 ± 0.003, with the smallest spread across seeds, consistent with the reduced over-estimation bias of its truncated distributional critics [21]. The gain is not specific to one weighting: with the score-aligned reward of carbon weight 1, TQC (M114) improves on SAC (M26) from 0.786 ± 0.008 to 0.774 ± 0.006 in Phase I and from 0.870 to 0.866 in score. With the history features (M112), the Phase I score (0.771 ± 0.007) is statistically indistinguishable from M27 (0.768 ± 0.005): cost and the grid KPI improve slightly (C 0.717, D 1.048) while emissions rise slightly (G 0.825). On the unseen buildings M112 is marginally better than M27 (public score 0.854 ± 0.006 vs. 0.859 ± 0.005, Table III), mainly through a lower D. Stacking the rich history with observation normalisation (M111) is clearly worse (0.795 ± 0.008). Combining the two best changes, TQC and the history features (M113), gives the best Phase I score of the study, 0.763 ± 0.005, through a lower cost (C 0.709) at unchanged emissions (G 0.816), and the largest savings on the unseen buildings (C 0.709, G 0.791); its score (0.862 ± 0.006) equals that of TQC alone. With TQC, the persistence forecast of carbon intensity and net load thus helps the policy to schedule cheaper charging, while with SAC its benefit was within the seed-to-seed spread.

TABLE V
FULL-BUDGET VARIANTS OF M27 (THREE SEEDS, MEAN ± STD; LOWER IS BETTER).

ID	Variant	Steps	Phase I	Score S	C	G
M102	M27 (short budget)	60k	0.794±.016	0.882±.006	0.751	0.838
M27	proposed (SAC)	200k	0.768±.005	0.867±.005	0.720	0.817
M112	+ history features	200k	0.771±.007	0.863±.004	0.717	0.825
M111	+ rich history + obs. norm.	200k	0.795±.008	0.891±.009	0.747	0.843
M109	M27 trained longer	400k	0.786±.013	0.885±.018	0.738	0.833
M110	TQC instead of SAC	200k	0.768±.004	**0.861**±.003	0.720	**0.816**
M113	TQC + history features	200k	**0.763**±.005	0.862±.006	**0.709**	**0.816**
M26	SAC, carbon weight 1	200k	0.786±.008	0.870±.006	0.732	0.840
M114	TQC, carbon weight 1	200k	0.774±.006	0.866±.004	0.721	0.826

Training length matters, but not monotonically. M27 scores 0.794 ± 0.016 after 60,000 steps, 0.768 ± 0.005 after 200,000 and 0.786 ± 0.013 when training continues to 400,000 steps from the 200,000-step checkpoints (M109): with a constant learning rate and entropy target, the policy drifts after it has converged, and the spread between seeds grows again. Early stopping on a validation year, or a decaying learning rate, would be needed to exploit longer training.

H. Comparison with the Challenge Leaderboard

The Phase I leaderboard of the 2022 challenge [8] ranked entries by $(C + G)/2$ on the same five buildings. Its best published entries scored 0.631, 0.635 and 0.651. Our best controller, M113 (0.763), would rank 11th among the 14 published entries, just ahead of the 11th and 12th entries (0.768, which M27 and M110 equal) and of the last two (0.800, 0.801). Leaderboard entries were free to use any technique, including hand-tuned rules, optimisation and forecasting; our result is obtained by a single model-free policy trained from reward alone. On the challenge's combined public score the best entry reached 0.728, compared with 0.853–0.854 for our TQC controllers (M110, M113), so a clear gap remains, mostly in the grid KPI D.

VII. DISCUSSION

What matters most. The ablations point to one principle: the reward should measure, on the right scale, what the evaluation measures, and as much of the uncontrollable variation as possible should be removed from it. SAC benefits more from these changes than PPO, and the gains add up.

Grid KPIs. All our best controllers increase ramping relative to no battery. Raising the ramping weight improves D (M26) but costs carbon and money, and the peak term is only a proxy

TABLE VI
UNITED NATIONS SUSTAINABLE DEVELOPMENT GOALS ADDRESSED BY THIS WORK.

SDG	Relevance	Justification
SDG 13 – Climate Action	★★★★★	Dynamic carbon-aware reward shaping minimises CO_2 emissions by adapting battery dispatch to real-time grid carbon intensity.
SDG 7 – Affordable and Clean Energy	★★★★★	Optimises battery storage and rooftop solar utilisation, improving energy efficiency and supporting renewable energy integration.
SDG 11 – Sustainable Cities and Communities	★★★★☆	Coordinates energy management across multiple residential buildings within the CityLearn smart-city environment.
SDG 9 – Industry, Innovation and Infrastructure	★★★☆☆	Demonstrates an AI-based control framework for intelligent energy infrastructure.
SDG 12 – Responsible Consumption and Production	★★☆☆☆	Encourages efficient electricity consumption and demand shifting toward lower-carbon periods.

for the load factor. A better per-step surrogate for $1 - L$ and ramp-aware action smoothing are promising directions.

Deployability. The controllers use only information available in a real building: present measurements, the environment's weather and tariff forecasts, and the measured battery power (for the counterfactual baseline during training). They transfer to unseen buildings without retraining, which supports their use as a starting point that can be fine-tuned on site.

Limitations. The study uses one climate, five training buildings and one year of data; carbon intensity and load are not forecast by the environment. Three seeds give reliable means at the full budget (standard deviations ≤ 0.008 for the best controllers), but at 60,000 steps the spread grows to 0.016–0.030, so single-seed screening at short budgets can mislead: the history features looked best with SAC on one seed, gave no reliable gain over three seeds, and only helped once combined with TQC at the full budget. Larger studies should use more seeds [32].

VIII. RELEVANCE TO THE SUSTAINABLE DEVELOPMENT GOALS

Battery control in grid-interactive buildings addresses several of the United Nations Sustainable Development Goals (SDGs). Table VI lists the goals this work contributes to, ordered by relevance.

IX. CONCLUSION

We studied 52 genuinely deployable deep RL controllers for community battery storage on the CityLearn 2022 benchmark. A rule-based controller and naive RL rewards were no better than leaving the batteries idle. A score-aligned counterfactual reward, whose per-step terms add up to the official KPIs on the no-battery scale and measure the improvement over idling, combined with SAC and an increased carbon weight, reduced electricity cost by 28.0% and emissions by 18.3% over three seeds. With TQC the same reward gave the best overall score (0.861), and TQC with past-only history features the best Phase I score (0.763; cost -29.1%, emissions -18.4%) and the largest savings on five unseen buildings (cost -29.1%, emissions -20.9%). Reward alignment, together with an off-policy learner, was the decisive factor; larger networks, CrossQ, multi-step returns and longer training did not help, and the choice between SAC and TQC and the history features changed the result by less than 1%. This gives practitioners a simple recipe for building RL controllers that optimise what operators are actually measured on.

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